MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

AGRICULTURE SCIENCE PAPER-I

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. India is situated in the:

(a) South-west trade wind zone (b) North-east trade wind zone

(c) East-west trade wind zone (d) Easterly trade wind zone

2. India's food grain production (2024-25) is estimated to be_____ million metric tonnes.

(a) 375 (b) 410

(c) 330 (d) 354

3. Pearl millet belongs to _____ plant.

(a) C_4 (b) C_3

(c) CAM (d) Both C_3 and C_4

4. International Potato Centre is located at:

(a) Nairobi, Kenya (b) Lima, Peru

(c) California, USA (d) Hyderabad, India

5. Tillage or ploughing 25-30 cm deep is:

(a) Medium deep ploughing (b) Medium ploughing

(c) Deep ploughing (d) Shallow ploughing

6. Sunflower is a native of:

(a) USA (b) Spain

(c) India (d) Africa

7. The earth is atmosphere roughly contain at present _____ % CO_2.

(a) 0.04 (b) 0.3

(c) 0.4 (d) 0.03

8. Mouldboard plough is a_____ implement.

(a) seed drill (b) planking

(c) primary tillage (d) secondary tillage

9. North eastern states comes under _____ agro climatic zone.

(a) Eastern plateau and hill region (b) Eastern Himalayan region

(c) North eastern Region (d) Western Himalayan region

10. Insect *Dactylopius tomentosus* is used as biological control of weeds for eradication of:

(a) Johnson grass (b) Congress grass

(c) Prickly pear (d) Lantana camara

11. Soybean seed contain _____ % protein.

(a) 15 (b) 30

(c) 20 (d) 40

12. The relative land area under sole crops that is required to produce the yields achieved in intercropping:

(a) Land equivalent ratio (b) Crop equivalent yields

(c) Relative yield total (d) Multiple cropping index

13. Low pressure area near the equator are called:

(a) Equinox (b) Polar calms

(c) Burst (d) Doldrums

14. Most common weed in rice.

(a) *Cyperus spp* (b) *Vicoa spp*

(c) *Echinochloa spp* (d) *Celosia spp*

15. ICAR Research Complex for Easter Region is located at:

(a) Barapani (b) Orissa

(c) Patna (d) Ranchi

16. Pendimethalin is a common pre emergence herbicide used in:

(a) Rice (b) Wheat

(c) Maize (d) Mustard

17. Orobanche is a:

(a) Shoot parasite (b) Root parasite

(c) Stem parasite (d) Flower parasite

18. Bajra (Pearl millet) generally contain _____ protein.

(a) 20% (b) 10.5%

(c) 12.5% (d) 11.6%

19. *Zea mays saccharata* belongs to:

(a) Sweet corn (b) Flint corn

(c) Pop corn (d) Dent corn

20. *Echinochloa crusgalli* is a common weed of:

(a) Soybean (b) Rice

(c) Mustard (d) Maize

21. Acidic soils are generally deficient in:

(a) P (b) Fe

(c) Cu (d) Al

22. Frankia is a _____N fixer.

(a) Free living (b) Associative

(c) Symbiotic (d) Non symbiotic

23. Leaf color chart is used as a diagnostic tool for:

(a) K (b) N

(c) S (d) P

24. Micro nutrient elements are:

(a) Ni and K (b) N and Cu

(c) Zn and S (d) Ni and Zn

25. Kaolinite is a _____ secondary mineral.

(a) 2:2 (b) 3:1

(c) 1:1 (d) 2:1

26. In India, Nitrogen is considered deficient, if soil contain less than _____ kg per ha.

(a) 180 (b) 200

(c) 280 (d) 560

27. _____ fertilizer is given as top dressing

(a) P (b) N

(c) S (d) K

28. According to USDA soil texture classification, silt is _____ mm.

(a) 0.05 to 0.1 (b) 0.002 to 0.02

(c) < 0.002 (d) 0.002 to 0.05

29. A silicate clay enriched horizon formed by illuviation of clay.

(a) Umbric horizon (b) Argillic horizon

(c) Oxic horizon (d) Agric horizon

30. Alfisol have more than _____ % base saturation.

(a) 10 (b) 50

(c) 35 (d) 40

31. Nitrogen deficiency in plant leads to:

(a) Yellowing of older leaves (b) Browning of tender leaves

(c) Purple colour of older leaves (d) Yellowing of tender elaves

32. Beer lambart law is applied in the analysis of:

(a) P (b) N

(c) Ca (d) pH

33. The final product of nitrification.

(a) NO_2 (b) NO_3

(c) NH (d) Elemental N

34. The diameter of potassium is _____ Å.

(a) 1.6 (b) 2.86

(c) 2.66 (d) 2

35. Example of straight fertilizer.

(a) Single super phosphate (b) Diammonium phosphate

(c) Ammonium potassium phosphate (d) Soluble NPK

36. Largest area under organic farming in India.

(a) Madhya Pradesh (b) Assam

(c) Sikkim (d) Uttarakhand

37. _____ is allowed to be used in organic farming.

(a) Urea (b) Rock phosphate

(c) NPK (d) DAP

38. The Mizoram Organic Farming Bill/Act was passed in the year.

(a) 2004 (b) 2000

(c) 2015 (d) 2010

39. The first Indian state to become fully organic state in 2016.

(a) Meghalaya (b) Goa

(c) Nagaland (d) Sikkim

40. Organic farming certification in India is governed by:

(a) RKVY (Rashtriya Krishi Vikas Yojana)

(b) NCIG (National Certification for Organic Product in India)

(c) NPOP (National Programme for Organic Production)

(d) SCAOF (Seed Certification Agency for Organic Farming)

41. Foundation seed is labelled in _____ colour.

(a) White (b) Blue

(c) Green (d) Golden yellow

42. _____ is used to increase panicle exertion of CMS lines in hybrid rice.

(a) Nitrogen (b) Gibberellic acid

(c) Cytokinin (d) Ethylene

43. Seed _____ is a pre sowing hydration treatment of seeds for rapid and uniform emergence, germination and growth.

(a) princhíng (b) blending

(c) vigour test (d) priming

44. _____ makes the seed more uniform in size, weight and shape.

(a) Pelleting (b) Priming

(c) Roughing (d) Blenching

45. The Seed Act was enacted by the parliament on:

(a) 29 December, 1986 (b) 29 December, 1996

(c) 29 December, 1966 (d) 29 December, 1976

46. In the universal soil loss equation, the K factor is :

(a) Crop management

(b) Slope

(c) ratio of soil loss with a specific erosion control

(d) Soil erodibility index

47. _____ is the practice of growing protective crops and row crops in alternative strips on the contours.

(a) Strip cropping (b) Contour cropping

(c) Selective cropping (d) Relay cropping

48. Under the land capability classes, Class II is denoted by:

(a) Red (b) Green

(c) Yellow (d) Black

49. The design discharge through the drainage channel is computed from the formula. Q = _____CIA

(a) 0.02 (b) 0.0028

(c) 0.028 (d) 0.035

50. Maximum (about 90% of soil movement) in wind erosion is below the height of:

(a) 30 cm (b) 10 cm

(c) 20 cm (d) 15 cm

SECTION - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **any five (5)** questions.*

1. Write the origin, botanical description, seed rate, spacing, method & time of sowing, cropping sequence, manure and fertilizers, common herbicides, harvesting, threshing and yield of Rice or Maize. Write the origin, botanical description, seed rate, spacing, method & time, cropping sequence, manure and fertilizers, harvesting and yield of Soybean or Rape Seed. **(12 + 8=20)**

2. Write notes on any four (4) of the following: **(5+5+5+5=20)**

(a) Cropping systems and types
(b) Conservation agriculture
(c) Secondary tillage and implements
(d) Integrated farming system (IFS)
(e) Weather forecasting
(f) Precision farming

3. Describe the principles of integrated weed management, herbicide selectivity with examples and herbicide formulations. Classify herbicides based on mode of action. **(6+5+6+3=20)**

4. Describe any four (4) of the following: **(5+5+5+5=20)**

(a) Integrated nutrient management
(b) Stokes law
(c) Types of soil structure
(d) C:N ratio and its significance to soil health
(e) Nursery management for Rice

5. What is the function of Nitrogen as an essential element and describe its deficiency symptoms? Describe N mineralization and factors affecting mineralization. How N can be organically managed in soils of Mizoram? **(6+9+5=20)**

6. Write the principles, key practice, benefits and challenges of organic farming. Describe in brief pest, disease and weed management in organic farming. Name and discuss any default organic traditional farming system practiced in Mizoram and how the P deficiency can be managed in the acidic soils. **(10+6+4=20)**

7. Define seed and discuss the different classes of seed. What is seed enhancement and what are the methods? Describe in brief Intellectual Property Rights. **(8+8+4=20)**

8. Define soil erosion and factors affecting soil erosion. Describe the different measures of soil conservation. What water conservation measures can be adopted in Mizoram to increase cropping intensity during *rabi* season? **(5+12+3=20)**

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